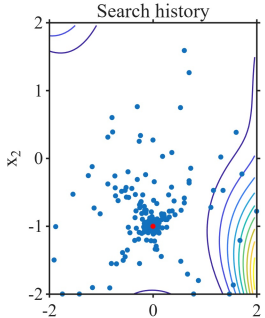
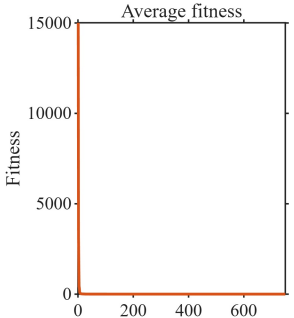
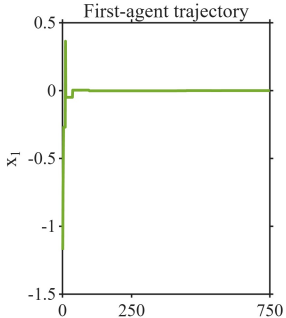
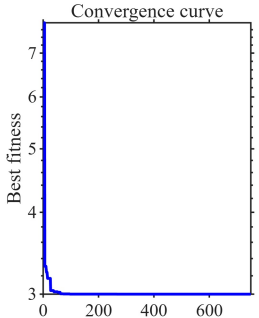

F18
×10^8
2.5
2
1.5
1
0.5
Fitness
4
2
0
-2
-4
x2
-4
-2
0
2
4
x1


Search history
2
1
0
-1
-2
x2
-2
0
2
x1


Average fitness
15000
10000
5000
0
Fitness
0
200
400
600
Iteration


First-agent trajectory
0.5
0
-0.5
-1
-1.5
x1
0
250
750
Iteration


Convergence curve
7
6
5
4
3
Best fitness
0
200
400
600
Iteration